

DAY 5 PRACTICE – PYTHON

Exercise 1 – String Length Without len()

```
...  
# Exercise 1 - String Length Without len()  
text = "dataengineering"  
  
length = 0  
for i in text:  
    length += 1  
print(f"Length of string: {length}")  
[1] ✓ 0.0s  
... Length of string: 15
```

Exercise 2 – Convert Sentence to Uppercase

```
...  
# Exercise 2 - Convert Sentence to Uppercase  
sentence = input("Enter a sentence: ")  
print(f"Input: {sentence}")  
  
print(f"Output: {sentence.upper()}")  
[2] ✓ 11.4s  
.. Input: data engineering is fun  
Output: DATA ENGINEERING IS FUN
```

Exercise 3 – Count Occurrence of a Character

```
...  
  
# Exercise 3 - Count Occurrence of a Character  
words = input("Enter a word: ")  
print(f"Word: {words}")  
char = input("Enter a character: ")  
  
count = 0  
for i in words:  
    if char == i:  
        count += 1  
  
print(f"count of \'{char}\' character in the word: {count}")
```

4] ✓ 6.0s

- Word: banana
count of 'a' character in the word: 3

Exercise 4 – Reverse a List

```
# Exercise 4 - Reverse a List  
  
numbers = [5,10,15,20,25]  
print(f"Reverse list is: {numbers[::-1]}")
```

✓ 0.0s

Reverse list is: [25, 20, 15, 10, 5]

Exercise 5 – Remove Duplicates From List

```
# Exercise 5 - Remove Duplicates From List  
numbers = [10,20,20,30,40,40,50]  
  
print(f"Unique list is: {list(set(numbers))}")
```

✓ 0.0s

Unique list is: [40, 10, 50, 20, 30]

Exercise 7 – Find Student With Highest Marks

```
'''  
# Exercise 7 - Find Student With Highest Marks  
marks = {  
    "Rahul": 85,  
    "Amit": 72,  
    "Priya": 90,  
    "Neha": 78  
}  
highest_mark = max(marks.values())  
for key, value in marks.items():  
    if value == highest_mark:  
        print(f"{key} scored of highest marks: {highest_mark}")
```

[4] ✓ 0.0s

.. Priya scored of highest marks: 90

Exercise 8 – Function to Count Words

```
'''  
# Exercise 8 - Function to Count Words  
  
def count_word(sentence):  
    n = sentence.split()  
    print(f"Number of words is: {len(n)}")  
  
count_word("Python is widely used in data engineering")
```

✓ 0.0s

Number of words is: 7

Exercise 9 – Read File and Print Line Numbers

```
...  
# Exercise 9 - Read File and Print Line Numbers  
with open("notes.txt", "w") as file:  
    file.write("Python is a popular programming language.\n")  
    file.write("It is easy to learn and use.\n")  
    file.write("Python is widely used in data analysis and automation.\n")  
    file.write("It has many libraries like Pandas and NumPy.\n")  
    file.write("Python is commonly used by data engineers and developers.")  
  
with open("notes.txt", "r") as file:  
    line_number = 1  
    for lines in file:  
        print(line_number, lines)  
        line_number += 1
```

✓ 0.0s

- 1 Python is a popular programming language.
- 2 It is easy to learn and use.
- 3 Python is widely used in data analysis and automation.
- 4 It has many libraries like Pandas and NumPy.
- 5 Python is commonly used by data engineers and developers.

Exercise 10 – Filter Data From File

```
...  
# Exercise 10 - Filter Data From File  
with open("scores.txt", "w") as file:  
    file.write("Rahul 85\n")  
    file.write("Amit 60\n")  
    file.write("Priya 92\n")  
    file.write("Neha 55\n")  
    file.write("Karan 75\n")  
  
with open("scores.txt", "r") as file:  
    data = file.readlines()  
    for i in data:  
        data1 = i.split()  
        name = data1[0]  
        marks = int(data1[1])  
  
        if marks > 70:  
            print(name, marks)
```

3] ✓ 0.0s

```
Rahul 85  
Priya 92  
Karan 75
```

Exercise 11 – CSV File Reading

```
# Exercise 11 - CSV File Reading

import csv

with open("employees.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Id", "Name", "Salary"])
    writer.writerow([1, "Rahul", 35000])
    writer.writerow([2, "Sneha", 42000])
    writer.writerow([3, "Amit", 60000])
    writer.writerow([4, "Priya", 38000])
    writer.writerow([5, "Rohit", 41000])
    writer.writerow([6, "Anjali", 65000])
    writer.writerow([7, "Karan", 36000])
    writer.writerow([8, "Meena", 43000])
    writer.writerow([9, "Vijay", 70000])
    writer.writerow([10, "Pooja", 39000])

with open("employees.csv", "r") as file:
    data = csv.reader(file)
    for row in data:
        print(row)
```

✓ 0.0s

```
['Id', 'Name', 'Salary']
['1', 'Rahul', '35000']
['2', 'Sneha', '42000']
['3', 'Amit', '60000']
['4', 'Priya', '38000']
['5', 'Rohit', '41000']
```

Exercise 12 – Simple Data Cleaning

```
# Exercise 12 - Simple Data Cleaning
data = ["Rahul", "Amit", "", "Priya", "", "Neha"]

clean_list = []
for i in data:
    if i != "":
        clean_list.append(i)
print(f"Clean list is: {clean_list}")
```

✓ 0.0s

```
Clean list is: ['Rahul', 'Amit', 'Priya', 'Neha']
```

SQL PRACTICE – DAY 5

-- 1. Display all students.

[SQL Worksheet]*      Aa 

```
1  -- 1. Display all students.
2  SELECT * FROM STUDENTS1;
```

Query result Script output DBMS output Explain Plan SQL history

  Download  Execution time: 0.009 seconds

	ID	NAME	CITY	MARKS
1	1	Rahul	Mumbai	85
2	2	Amit	Pune	72
3	3	Priya	Delhi	90
4	4	Neha	Bangalore	78
5	5	Karan	Chennai	65
6	6	Sneha	Hyderabad	88
7	7	Anita	Kolkata	74

-- 2. Display students who scored more than 80 marks.

```
3
4  -- 2. Display students who scored more than 80 marks.
5  SELECT NAME, MARKS FROM STUDENTS1 WHERE MARKS > 80;
```

Query result Script output DBMS output Explain Plan SQL history

  Download  Execution time: 0.003 seconds

	NAME	MARKS
1	Rahul	85
2	Priya	90
3	Sneha	88
4	Pooja	92
5	Anita	81

-- 3.Display students whose city is "Pune".

```
6
7  -- 3.Display students whose city is "Pune".
8  SELECT * FROM STUDENTS1 WHERE CITY = 'Pune';
```

Query result Script output DBMS output Explain Plan SQL history				
		Download ▾	Execution time: 0.004 seconds	
	ID	NAME	CITY	MARKS
1	2	Amit	Pune	72
2	10	Anita	Pune	81

--4.Find average marks of all students.

[SQL Worksheet]* ▾      Aa ▾ 

```
1  --4.Find average marks of all students.
2  SELECT AVG(MARKS) AS AVERAGE_MARKS FROM STUDENTS1;
3
```

Query result Script output DBMS output Explain Plan SQL his	
	 Download ▾ Execution time: 0.004 seconds
	AVERAGE_MARKS
1	79.4

-- 5.Count number of students in the table.

[SQL Worksheet]*

Aa

1

-- 5.Count number of students in the table.

2

SELECT COUNT (*) AS STUDENT_COUNT FROM STUDENTS1;

3

4

Query result

Script output

DBMS output

Explain Plan

i

Download

Execution time: 0.002 seconds

	STUDENT_COUNT
1	10

-- 6.Display students sorted by marks in descending order

[SQL Worksheet]*

Aa

1

2

-- 6.Display students sorted by marks in descending order.

3

SELECT * FROM STUDENTS1 ORDER BY MARKS DESC;

4

-- 7.Display distinct cities.

5

Query result

Script output

DBMS output

Explain Plan

SQL history

i

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Execution time: 0.004 seconds

	ID	NAME	CITY	MARKS
1	8	Pooja	Jaipur	92
2	3	Priya	Delhi	90
3	6	Sneha	Hyderabad	88
4	1	Rahul	Mumbai	85
5	10	Anita	Pune	81

-- 7.Display distinct cities.

```
3  SELECT * FROM STUDENTS1 ORDER BY MARKS DESC;  
4  -- 7.Display distinct cities.  
5  SELECT DISTINCT CITY FROM STUDENTS1;  
6
```

Query result

Script output

DBMS output

Explain Plan



Download



Execution time: 0.004 seconds

	CITY
1	Pune
2	Chennai
3	Hyderabad
4	Jaipur
5	Kolkata

LINUX PRACTICE – DAY 5

1. Create a directory called data_lab.
2. Inside the directory create files:
students.txt, marks.txt, data.txt
3. List all files inside the directory.

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ mkdir -p data_lab
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ ls -lrth
total 4.0K
drwxrwxrwx 1 prashant prashant 4.0K Mar  6 20:20 linux_exercise
-rwxrwxrwx 1 prashant prashant 876 Mar  6 20:25 numbers.txt
drwxrwxrwx 1 prashant prashant 4.0K Mar  6 20:42 logs
drwxrwxrwx 1 prashant prashant 4.0K Mar  7 17:32 data_files
drwxrwxrwx 1 prashant prashant 4.0K Mar  8 17:27 project_data
drwxrwxrwx 1 prashant prashant 4.0K Mar  9 2026 data_lab
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise$ cd data_lab/
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ touch students.txt marks.txt data.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ ls -lrth
total 0
-rwxrwxrwx 1 prashant prashant 0 Mar  9 22:59 students.txt
-rwxrwxrwx 1 prashant prashant 0 Mar  9 22:59 marks.txt
-rwxrwxrwx 1 prashant prashant 0 Mar  9 22:59 data.txt
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ |
```

4. Display the first 5 lines of students.txt.
5. Display the last 3 lines of marks.txt.
6. Count number of lines in data.txt.
7. Search the word “Python” inside students.txt.

```
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ head -n 5 students.txt
id name city subject
1 Rahul Mumbai Python
2 Amit Pune SQL
3 Priya Delhi Linux
4 Neha Bangalore Python
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ tail -n 3 marks.txt
8 Linux 92
9 DataEngineering 69
10 SQL 81
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ cat data.txt | wc -l
11
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ grep -i "PYthoN" students.txt
1 Rahul Mumbai Python
4 Neha Bangalore Python
6 Sneha Hyderabad Python
prashant@DESKTOP-B9KE2N7:/mnt/c/Users/admin/practice/linux_exercise/data_lab$ |
```